

INVITED TALK / ABSTRACT

Future stability of Maxwell-Jüttner equilibria and vacuum for the massless Boltzmann equation on FLRW spacetimes

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Abstract

In this talk, we will consider the massless Boltzmann equation on FLRW spacetimes with spatial topology $\mathbb{T}^3$ in the linear and decelerated expanding regimes. In these backgrounds, the massless Boltzmann equation admits a non-stationary Maxwell-Jüttner equilibrium that maximises the entropy of the system. In this work, this equilibrium is shown to be non-linearly stable for the massless Boltzmann equation in the case of hard ball interaction. We will also discuss the non-linear stability of the vacuum solution for sufficiently fast expanding spacetimes. This work is motivated by the stability of solutions to the Einstein–massless Boltzmann system. This is joint work with Robert M. Strain (University of Pennsylvania) and Martin Taylor (Imperial College London).

Keywords: massless Boltzmann equation; FLRW spacetimes; Maxwell–Jüttner equilibrium; future stability; non-linear stability; vacuum solution; hard ball interaction; expanding spacetimes; Einstein–massless Boltzmann system